Core Strategy

MQL5 — Extensible Multi-Strategy Trading EA

1. DOCUMENT PURPOSE

Dokumen ini merupakan technical specification untuk membangun Expert Advisor menggunakan:

```
Platform  : MetaTrader 5
Language  : MQL5
Timeframe : H1 + M15
```

Dokumen harus cukup deterministic sehingga developer dapat mengimplementasikan EA tanpa perlu membuat asumsi baru terhadap trading rules.

Dokumen juga mendefinisikan architecture agar strategy baru dapat ditambahkan tanpa mengubah core trading engine.

2. PRIMARY OBJECTIVES

EA harus memenuhi:

```
1. Deterministic
2. Non-repainting
3. No lookahead bias
4. Risk controlled
5. Testable
6. Maintainable
7. Extensible
8. Broker-aware
9. Strategy-independent execution
```

Prioritas:

```
Correctness
    ↓
Risk / Security
```

```
    ↓
Maintainability
    ↓
Readability
    ↓
Performance
    ↓
Extensibility
```

---

3. DESIGN PRINCIPLE

Architecture menggunakan prinsip:

```
Strategy
    ↓
TradeSignal
    ↓
Guardian
    ↓
RiskEngine
    ↓
TradeExecutor
```

Strategy tidak boleh melakukan order secara langsung.

Contoh yang DILARANG:

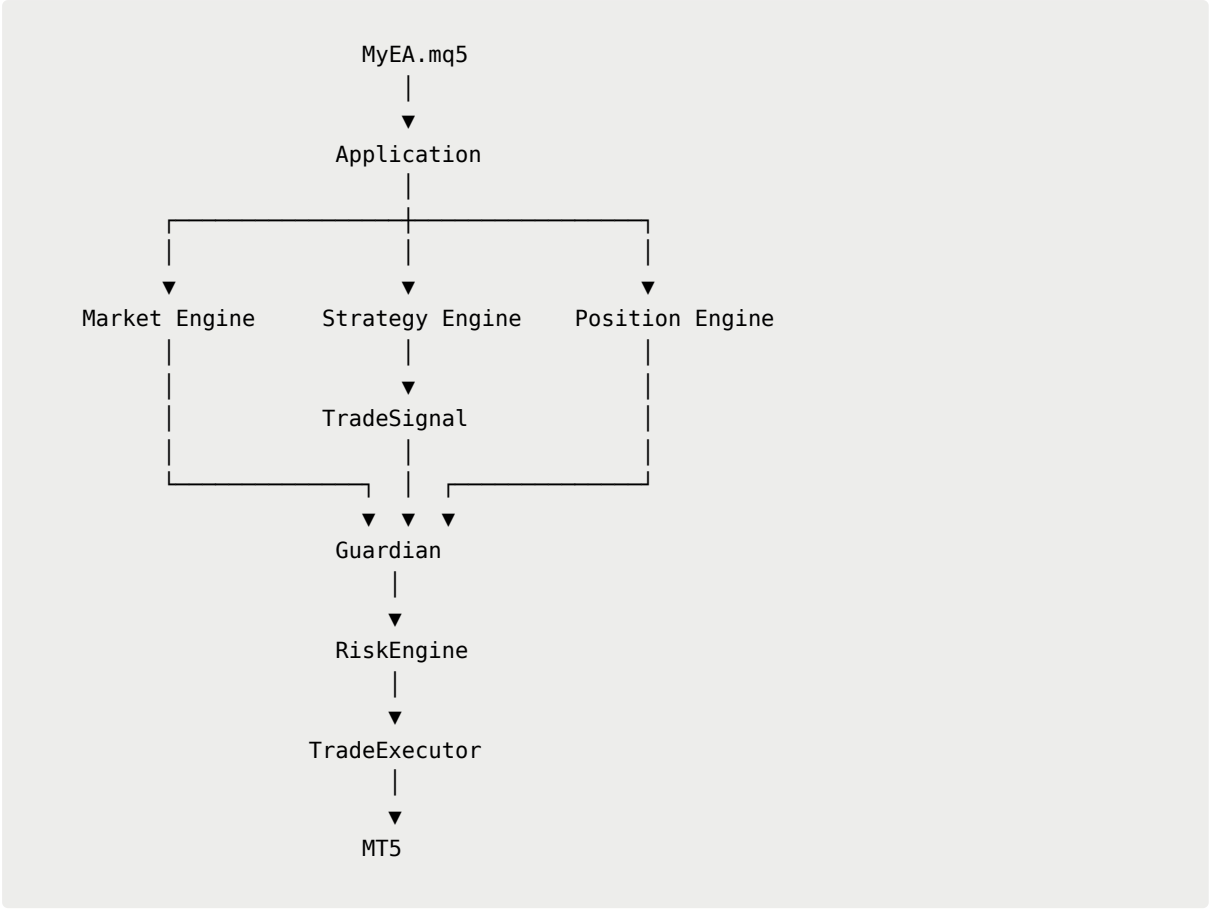
```
RangeStrategy
    ↓
CTrade.Buy()
```

Yang benar:

```
RangeStrategy
    ↓
TradeSignal
    ↓
Guardian
    ↓
RiskEngine
    ↓
TradeExecutor
    ↓
CTrade
```

---

4. HIGH LEVEL ARCHITECTURE



5. MODULE STRUCTURE

Recommended project structure:

```
Experts/
└── TradingEA/
    │
    ├── TradingEA.mq5
    │
    ├── Core/
    │   ├── Types.mqh
    │   ├── Config.mqh
    │   ├── MarketContext.mqh
    │   ├── TradeSignal.mqh
    │   └── EAState.mqh
    │
    ├── Market/
    │   ├── MarketData.mqh
    │   ├── IndicatorManager.mqh
    │   ├── RegimeEngine.mqh
```

```
│   └── MarketStructure.mqh
│
├── Strategies/
│   ├── IStrategy.mqh
│   ├── StrategyEngine.mqh
│   ├── RangeStrategy.mqh
│   ├── BreakoutStrategy.mqh
│   └── TrendPullbackStrategy.mqh
│
├── Risk/
│   ├── RiskEngine.mqh
│   └── Guardian.mqh
│
├── Trading/
│   ├── TradeExecutor.mqh
│   ├── PortfolioManager.mqh
│   └── PositionManager.mqh
│
└── Utils/
    ├── PriceUtils.mqh
    ├── VolumeUtils.mqh
    ├── TimeUtils.mqh
    └── Logger.mqh
```

Architecture ini adalah modular tetapi bukan Clean Architecture penuh.

Tidak diperlukan:

```
Repository
DTO framework
Dependency Injection framework
Service Locator
Factory framework
Event Bus
```

kecuali kebutuhan tersebut muncul pada future version.

---

6. CORE ENUMS

EA menggunakan enum untuk menghindari string-based state.

Direction

```
SIGNAL_NONE
SIGNAL_BUY
SIGNAL_SELL
```

Regime

```
REGIME_SIDEWAYS
REGIME_TRANSITION
REGIME_TREND_BULLISH
REGIME_TREND_BEARISH
```

Strategy

```
STRATEGY_NONE
STRATEGY_RANGE
STRATEGY_BREAKOUT_RETEST
STRATEGY_TREND_PULLBACK
```

EA State

```
EA_RUNNING
EA_DISABLED
```

Breakout State

```
BREAKOUT_IDLE
BREAKOUT_WAITING_RETEST
```

---

7. TRADE SIGNAL CONTRACT

Semua strategy wajib menghasilkan object `TradeSignal`.

Minimal field:

```
valid
strategy
direction

signalTime

entryPrice

stopLoss
```

```
takeProfit

riskPercent
riskAmount
volume

reason
```

Strategy hanya mengisi:

```
strategy
direction
signalTime
entryPrice
stopLoss
takeProfit
reason
valid
```

RiskEngine mengisi:

```
riskPercent
riskAmount
volume
```

Strategy tidak menentukan final volume.

---

8. MARKET CONTEXT

Strategy tidak boleh mengambil data market secara acak dari berbagai module.

Semua data yang dibutuhkan strategy disediakan melalui `MarketContext`.

Minimal:

```
CurrentSymbol

CurrentBid
CurrentAsk
CurrentSpread

H1:
    Close
    EMA50
    EMA200
    ADX
```

```
    ATR
    NormalizedSlope

M15:
    OHLC
    EMA50
    ADX
    ATR

Regime

MarketStructure

RangeData
```

`MarketContext` hanya berisi snapshot data untuk satu evaluation cycle.

---

9. DATA OWNERSHIP

MarketData

Bertanggung jawab:

```
OHLC
Time
Bid
Ask
Spread
```

IndicatorManager

Bertanggung jawab:

```
EMA
ADX
ATR
Bollinger
```

RegimeEngine

Bertanggung jawab:

```
H1 Regime
```

MarketStructure

Bertanggung jawab:

```
Confirmed Swing High
Confirmed Swing Low
HH
HL
LH
LL
```

Strategy

Bertanggung jawab:

```
Setup detection
SL/TP structural calculation
TradeSignal
```

RiskEngine

Bertanggung jawab:

```
Risk
Volume
```

Guardian

Bertanggung jawab:

```
Trading permission
```

TradeExecutor

Bertanggung jawab:

```
Order execution
```

PositionManager

Bertanggung jawab:

```
Existing position management
BE
Partial close
```

---

10. TIMEFRAME RULE

H1:

```
Context / Regime
```

M15:

```
Setup / Entry
```

M5:

```
NOT USED
```

---

11. CANDLE INDEXING CONTRACT

Semua strategy menggunakan closed candle.

```
shift 0 = current / incomplete candle
shift 1 = latest closed candle
shift 2 = previous closed candle
```

Signal candle:

```
shift = 1
```

Current candle:

```
shift = 0
```

Tidak boleh digunakan untuk membuat trading decision.

---

12. NEW M15 BAR

Strategy evaluation hanya dilakukan satu kali pada candle M15 baru.

Pseudo:

```
OnTick():

    ManagePositions()

    if !IsNewM15Bar():
        return

    ProcessNewM15Bar()
```

`ProcessNewM15Bar()` menggunakan candle shift 1 sebagai signal candle.

---

13. INDICATOR CONFIGURATION

H1

```
EMA Fast = 50
EMA Slow = 200
ADX = 14
ATR = 14
```

M15

```
EMA = 50
ADX = 14
ATR = 14
Bollinger = 20 / 2
```

Bollinger tersedia tetapi tidak digunakan pada v2.0.

---

14. INDICATOR DATA VALIDATION

Sebelum strategy evaluation:

```
CopyBuffer()
```

harus berhasil.

Jika data tidak tersedia:

```
No signal
```

EA tidak boleh menggunakan nilai default `0` sebagai pengganti indicator data yang gagal.

---

15. H1 REGIME ENGINE

Regime:

```
TREND_BULLISH
TREND_BEARISH
SIDEWAYS
TRANSITION
```

---

16. SLOPE

Parameter:

```
SlopeLookback = 3
TrendSlopeThreshold = 0.05
```

Formula:

```
EMA50Slope =
EMA50[1] - EMA50[4]
```

Normalized:

```
NormalizedSlope =
(EMA50[1] - EMA50[4])
/
ATR14[1]
```

---

17. BULLISH TREND

```
Close[1] > EMA200[1]

AND

EMA50[1] > EMA200[1]

AND

NormalizedSlope >= 0.05

AND

ADX14[1] >= 20
```

Output:

```
REGIME_TREND_BULLISH
```

---

18. BEARISH TREND

```
Close[1] < EMA200[1]

AND

EMA50[1] < EMA200[1]

AND

NormalizedSlope <= -0.05

AND

ADX14[1] >= 20
```

Output:

```
REGIME_TREND_BEARISH
```

---

19. SIDEWAYS

Parameters:

```
SidewaysADX = 18
SidewaysSlopeThreshold = 0.05
```

Condition:

```
ADX14[1] < 18

AND

abs(NormalizedSlope) < 0.05
```

Output:

```
REGIME_SIDEWAYS
```

---

20. TRANSITION

Jika tidak memenuhi:

```
TREND_BULLISH
TREND_BEARISH
SIDEWAYS
```

maka:

```
REGIME_TRANSITION
```

Priority:

```
TREND
    ↓
SIDEWAYS
    ↓
TRANSITION
```

---

21. STRATEGY ROUTING

Regime menentukan strategy yang boleh aktif.

```
SIDEWAYS
    ↓
RangeStrategy

TRANSITION
    ↓
BreakoutRetestStrategy

TREND_BULLISH
    ↓
TrendPullbackStrategy BUY

TREND_BEARISH
    ↓
TrendPullbackStrategy SELL
```

Strategy lain tidak dievaluasi.

---

22. STRATEGY INTERFACE

Semua strategy mengikuti contract yang sama.

Concept:

```
IStrategy
    Evaluate(
        MarketContext,
        TradeSignal
    )
```

Return:

```
true
```

jika menghasilkan candidate signal.

Return:

```
false
```

jika tidak ada signal.

Strategy tidak boleh melakukan order.

---

23. RANGE PARAMETERS

```
RangeLookback = 20

MinRangeATR = 1.5
MaxRangeATR = 4.0

InteractionATR = 0.20

RangeZonePercent = 0.20

RejectionWickBody = 1.50
```

---

24. RANGE CALCULATION

Signal candle:

```
shift 1
```

Range:

```
shift 2 ... shift 21
```

Formula:

```
RangeHigh =
Highest High(2..21)

RangeLow =
Lowest Low(2..21)
```

Signal candle tidak boleh memengaruhi RangeHigh/RangeLow.

---

25. RANGE SIZE

```
RangeSize =
RangeHigh - RangeLow
```

Valid:

```
RangeSize > 0

AND

RangeSize >= ATR[1] × 1.5

AND

RangeSize <= ATR[1] × 4.0
```

---

26. RANGE INTERACTION

Tolerance:

```
InteractionTolerance =
ATR[1] × 0.20
```

Upper interaction:

```
High[i] >= RangeHigh - InteractionTolerance
```

Lower interaction:

```
Low[i] <= RangeLow + InteractionTolerance
```

Candle range:

```
i = 2..21
```

Minimum:

```
UpperInteractions >= 2
LowerInteractions >= 2
```

Satu candle hanya dihitung satu kali.

---

27. RANGE POSITION

```
RangePosition =
(Close[1] - RangeLow)
/
RangeSize
```

BUY:

```
RangePosition <= 0.20
```

SELL:

```
RangePosition >= 0.80
```

---

28. REJECTION CANDLE

Body:

```
Body =
abs(Close[1] - Open[1])
```

Jika:

```
Body == 0
```

maka rejection:

```
FALSE
```

Lower wick:

```
LowerWick =
min(Open[1], Close[1]) - Low[1]
```

Upper wick:

```
UpperWick =
High[1] - max(Open[1], Close[1])
```

---

29. BULLISH REJECTION

```
Close[1] > Open[1]

AND

LowerWick >= Body × 1.50
```

```
AND

Low[1] <= RangeLow + InteractionTolerance

AND

Close[1] > RangeLow
```

---

30. BEARISH REJECTION

```
Close[1] < Open[1]

AND

UpperWick >= Body × 1.50

AND

High[1] >= RangeHigh - InteractionTolerance

AND

Close[1] < RangeHigh
```

---

31. RANGE BUY SIGNAL

```
Regime == SIDEWAYS

AND

RangeValid

AND

RangePosition <= 0.20

AND

BullishRejection
```

Signal:

```
Direction = BUY

EntryPrice = current Ask

SL = RangeLow - ATR[1] × 0.20

TP1 = MidRange

TP2 = RangeHigh
```

Untuk standardisasi `TradeSignal`, RangeStrategy mengembalikan:

```
TakeProfit = TP2
```

TP1 disimpan sebagai position-management target.

---

32. RANGE SELL SIGNAL

```
Regime == SIDEWAYS

AND

RangeValid

AND

RangePosition >= 0.80

AND

BearishRejection
```

Signal:

```
Direction = SELL

EntryPrice = current Bid

SL = RangeHigh + ATR[1] × 0.20

TP = RangeLow
```

TP1:

```
MidRange
```

---

33. BREAKOUT PARAMETERS

```
BreakoutBodyATR = 0.50

BreakoutADX = 20

RetestATR = 0.20

MaxRetestBars = 8
```

---

34. BREAKOUT ATR EXPANSION

```
ATR[1] > ATR[2]
```

AND:

```
ATR_SMA5 =
average(
    ATR[2],
    ATR[3],
    ATR[4],
    ATR[5],
    ATR[6]
)
```

Condition:

```
ATR[1] >= ATR_SMA5
```

---

35. BREAKOUT ADX EXPANSION

```
ADX[1] > ADX[2]

AND

ADX[1] >= 20
```

---

36. BREAKOUT BUY

```
Regime == TRANSITION

AND

Close[1] > RangeHigh

AND

Body >= ATR[1] × 0.50

AND

ATR expansion

AND

ADX expansion

AND

Close[1] > High[2]
```

Jika valid:

```
BreakoutState = WAITING_RETEST
```

Store:

```
BreakoutDirection = BUY
BreakoutLevel = RangeHigh
BreakoutBarTime = Time[1]
BreakoutBarIndex = 1
```

---

37. BREAKOUT SELL

```
Regime == TRANSITION

AND

Close[1] < RangeLow

AND

Body >= ATR[1] × 0.50

AND

ATR expansion

AND

ADX expansion

AND

Close[1] < Low[2]
```

Store:

```
BreakoutDirection = SELL
BreakoutLevel = RangeLow
BreakoutBarTime = Time[1]
BreakoutBarIndex = 1
```

---

38. BREAKOUT STATE OWNERSHIP

Hanya:

```
BreakoutStrategy
```

yang boleh mengubah:

```
BreakoutState
BreakoutDirection
```

```
BreakoutLevel
BreakoutBarTime
```

Strategy lain tidak boleh mengubah state tersebut.

---

39. BREAKOUT RETEST

Tolerance:

```
RetestTolerance =
ATR[1] × 0.20
```

BUY:

```
Low[1] <= BreakoutLevel + RetestTolerance

AND

Close[1] >= BreakoutLevel
```

SELL:

```
High[1] >= BreakoutLevel - RetestTolerance

AND

Close[1] <= BreakoutLevel
```

---

40. RETEST CONFIRMATION

BUY:

```
RetestValid

AND

Close[1] > Open[1]
```

SELL:

```
RetestValid

AND

Close[1] < Open[1]
```

---

41. FAILED BREAKOUT

BUY:

```
BreakoutState == WAITING_RETEST

AND

Close[1] < BreakoutLevel
```

SELL:

```
BreakoutState == WAITING_RETEST

AND

Close[1] > BreakoutLevel
```

Action:

```
BreakoutState = IDLE
```

No trade.

---

42. RETEST EXPIRY

Maximum:

```
MaxRetestBars = 8
```

Jika:

```
CurrentM15BarCount
-
BreakoutBarCount
> 8
```

maka:

```
BreakoutState = IDLE
```

No trade.

---

43. BREAKOUT RETEST SIGNAL

BUY:

```
RetestValid
AND
BullishConfirmation
```

Signal:

```
Direction = BUY

EntryPrice = current Ask
```

SELL:

```
RetestValid
AND
BearishConfirmation
```

Signal:

```
Direction = SELL

EntryPrice = current Bid
```

---

44. CONFIRMED SWING

Pivot parameters:

```
SwingLeft = 2
SwingRight = 2
```

Swing Low:

```
Low[i] < Low[i-1]
Low[i] < Low[i-2]
Low[i] < Low[i+1]
Low[i] < Low[i+2]
```

Swing High:

```
High[i] > High[i-1]
High[i] > High[i-2]
High[i] > High[i+1]
High[i] > High[i+2]
```

Pivot hanya boleh digunakan setelah `SwingRight` candles telah closed.

---

45. BREAKOUT SL

BUY:

```
SL =
LatestConfirmedSwingLow
-
ATR[1] × 0.20
```

SELL:

```
SL =
LatestConfirmedSwingHigh
+
ATR[1] × 0.20
```

Fallback jika tidak tersedia:

BUY:

```
SL =
BreakoutLevel - ATR[1] × 0.20
```

SELL:

```
SL =
BreakoutLevel + ATR[1] × 0.20
```

---

46. BREAKOUT TP

```
RiskDistance =
abs(EntryPrice - SL)
```

```
TP =
EntryPrice ± RiskDistance × 2.0
```

BUY:

```
Entry + RiskDistance × 2
```

SELL:

```
Entry - RiskDistance × 2
```

---

47. MARKET STRUCTURE

MarketStructure module menyediakan:

```
LatestSwingHigh
PreviousSwingHigh
```

```
LatestSwingLow
PreviousSwingLow
```

---

48. BULLISH STRUCTURE

```
LatestSwingHigh > PreviousSwingHigh

AND

LatestSwingLow > PreviousSwingLow
```

Output:

```
BULLISH_STRUCTURE
```

---

49. BEARISH STRUCTURE

```
LatestSwingHigh < PreviousSwingHigh

AND

LatestSwingLow < PreviousSwingLow
```

Output:

```
BEARISH_STRUCTURE
```

---

50. TREND PULLBACK PARAMETERS

```
PullbackMinATR = 0.30

PullbackZoneATR = 0.30

SLBufferATR = 0.20
```

```
TrendRR = 2.0
```

51. BULLISH PULLBACK DEFINITION

Condition:

```
Regime == TREND_BULLISH

AND

M15 Structure == BULLISH
```

Latest bullish impulse:

```
Latest confirmed Swing High
```

Pullback mulai setelah latest swing high.

Pullback distance:

```
PullbackMove =
LatestSwingHigh - CurrentRelevantLow
```

Valid:

```
PullbackMove >= ATR[1] × 0.30
```

`CurrentRelevantLow` adalah lowest low sejak latest confirmed swing high sampai signal candle.

52. BEARISH PULLBACK DEFINITION

Condition:

```
Regime == TREND_BEARISH

AND
```

```
M15 Structure == BEARISH
```

Latest bearish impulse:

```
Latest confirmed Swing Low
```

Pullback:

```
PullbackMove =
CurrentRelevantHigh - LatestSwingLow
```

Valid:

```
PullbackMove >= ATR[1] × 0.30
```

`CurrentRelevantHigh` adalah highest high sejak latest confirmed swing low sampai signal candle.

---

53. EMA50 PULLBACK ZONE BUY

BUY zone:

```
Low[1] <= EMA50[1]
AND
Close[1] >= EMA50[1]
```

OR:

```
abs(Low[1] - EMA50[1])
<= ATR[1] × 0.30
```

---

54. EMA50 PULLBACK ZONE SELL

SELL zone:

```
High[1] >= EMA50[1]
AND
Close[1] <= EMA50[1]
```

OR:

```
abs(High[1] - EMA50[1])
<= ATR[1] × 0.30
```

---

55. TREND BUY SIGNAL

```
Regime == TREND_BULLISH

AND

BullishStructure

AND

ValidBullishPullback

AND

EMA50Zone

AND

Close[1] > Open[1]

AND

Close[1] > High[2]
```

Signal:

```
Direction = BUY
EntryPrice = current Ask
```

---

56. TREND SELL SIGNAL

```
Regime == TREND_BEARISH

AND

BearishStructure

AND

ValidBearishPullback

AND

EMA50Zone

AND

Close[1] < Open[1]

AND

Close[1] < Low[2]
```

Signal:

```
Direction = SELL
EntryPrice = current Bid
```

---

57. TREND SL

BUY:

```
SL =
PullbackSwingLow
-
ATR[1] × 0.20
```

SELL:

```
SL =
PullbackSwingHigh
+
ATR[1] × 0.20
```

Jika structural swing tidak tersedia:

```
No Signal
```

Tidak menggunakan arbitrary fallback untuk Trend Pullback.

---

58. TREND TP

```
RiskDistance =
abs(EntryPrice - SL)
```

BUY:

```
TP =
EntryPrice + RiskDistance × 2
```

SELL:

```
TP =
EntryPrice - RiskDistance × 2
```

---

59. RISK ENGINE

Default:

```
RiskPerTrade = 0.50%
MaxRiskPerTrade = 0.50%
```

Risk amount:

```
RiskAmount =
AccountEquity × 0.50%
```

---

60. POSITION SIZE

EA membaca:

```
SYMBOL_TRADE_TICK_SIZE
SYMBOL_TRADE_TICK_VALUE
SYMBOL_VOLUME_MIN
SYMBOL_VOLUME_MAX
SYMBOL_VOLUME_STEP
```

Risk per lot:

```
RiskPerLot =
PriceDistance / TickSize × TickValue
```

Volume:

```
Volume =
RiskAmount / RiskPerLot
```

Volume dibulatkan DOWN ke `SYMBOL_VOLUME_STEP`.

Tujuannya memastikan actual risk tidak melebihi configured risk.

---

61. VOLUME VALIDATION

Jika:

```
Volume < SYMBOL_VOLUME_MIN
```

maka:

```
REJECT
```

EA tidak menaikkan volume secara paksa ke minimum broker.

Jika:

```
Volume > SYMBOL_VOLUME_MAX
```

volume dibatasi ke maximum broker.

Setelah normalization:

```
ActualRisk <= MaxRiskPerTrade
```

harus tetap benar.

---

62. ACCOUNT RISK SOURCE

Default:

```
AccountEquity
```

bukan balance.

Risk calculation:

```
RiskAmount =
Equity × RiskPercent
```

---

63. GUARDIAN

Guardian adalah gate sebelum RiskEngine dan execution.

Checks:

```
TradingEnabled
Spread
Session
DailyLoss
Drawdown
OpenPosition
Symbol
Signal
```

```
SL
TP
```

Semua harus PASS.

---

64. MAX OPEN POSITION

v2.0:

```
MaxTotalOpenPositions = 1
```

EA hanya boleh mempunyai satu position aktif yang dikelola EA.

Untuk future version parameter ini dapat dikembangkan.

---

65. POSITION OWNERSHIP

EA hanya mengelola position yang memenuhi:

```
Symbol == configured symbol

AND

MagicNumber == EA MagicNumber
```

EA tidak boleh memodifikasi position dari:

```
Manual trade
EA lain
Magic number lain
```

---

66. MAGIC NUMBER

Input:

```
MagicNumber
```

Magic number harus unique untuk EA instance.

67. SPREAD GUARD

Input:

```
MaxSpreadPoints
```

Current spread:

```
Ask - Bid
```

Jika:

```
SpreadPoints > MaxSpreadPoints
```

maka:

```
REJECT ENTRY
```

Position management tetap berjalan.

68. SESSION FILTER

Inputs:

```
UseSessionFilter
SessionStart
SessionEnd
```

Jika disabled:

```
No session restriction
```

Jika enabled:

```
Entry hanya diperbolehkan dalam configured session
```

Time basis:

```
Broker server time
```

---

69. DAILY LOSS LIMIT

Default:

```
DailyLossLimit = 2%
```

Daily loss dihitung berdasarkan equity baseline awal hari.

```
DailyLossPercent =
(StartOfDayEquity - CurrentEquity)
/
StartOfDayEquity × 100
```

Jika:

```
DailyLossPercent >= 2%
```

maka:

```
EA = DISABLED_FOR_NEW_ENTRIES
```

Existing position tidak dipaksa close.

Reset pada broker server date berikutnya.

---

70. MAX DRAWDOWN

Default:

```
MaxDrawdown = 10%
```

Peak equity:

```
PeakEquity
```

Drawdown:

```
Drawdown =
(PeakEquity - CurrentEquity)
/
PeakEquity × 100
```

Jika:

```
Drawdown >= 10%
```

maka:

```
EA = DISABLED
```

Reset drawdown protection dilakukan manual.

---

71. BROKER VALIDATION

Sebelum execution, EA harus memvalidasi:

```
TickSize
TickValue
VolumeMin
VolumeMax
VolumeStep
StopsLevel
FreezeLevel
Digits
Point
```

EA tidak boleh hardcode:

```
5 digits
0.00001 point
0.01 volume step
```

---

72. PRICE NORMALIZATION

Semua:

```
Entry
SL
TP
```

harus dinormalisasi menggunakan symbol specification.

Price harus valid terhadap:

```
TickSize
Digits
```

---

73. STOP LEVEL

EA membaca:

```
SYMBOL_TRADE_STOPS_LEVEL
```

Jika SL/TP terlalu dekat:

```
REJECT
```

EA tidak boleh mengubah SL/TP secara arbitrary hanya agar broker menerima order.

---

74. FREEZE LEVEL

Jika broker memiliki:

```
SYMBOL_TRADE_FREEZE_LEVEL
```

Position modification harus memperhatikan freeze level.

Jika modification tidak valid:

```
Log
No forced modification
```

---

75. TRADE EXECUTOR

TradeExecutor adalah satu-satunya module yang bertanggung jawab mengirim order.

Implementasi menggunakan:

```
CTrade
```

Strategy tidak boleh menggunakan `CTrade`.

---

76. EXECUTION FLOW

```
TradeSignal
    ↓
Guardian
    ↓
RiskEngine
    ↓
Final Validation
    ↓
TradeExecutor
    ↓
CTrade
    ↓
Trade Result
```

---

77. EXECUTION PRICE

BUY:

```
Ask
```

SELL:

```
Bid
```

Harga aktual broker menjadi execution price.

RiskEngine harus menghitung risk berdasarkan actual intended entry price.

Setelah execution, EA harus membaca actual position price untuk management selanjutnya.

---

78. EXECUTION FAILURE

Setelah `CTrade`:

```
Check ResultRetcode()
```

Jika gagal:

```
Log:
retcode
description
symbol
direction
volume
SL
TP
```

Tidak boleh infinite retry.

Maximum:

```
1 retry
```

Retry hanya untuk temporary/recoverable errors.

---

79. POSITION MANAGER

PositionManager berjalan setiap tick.

Tanggung jawab:

```
Detect position
Manage BE
Manage partial close
Detect closed position
```

PositionManager tidak mencari entry baru.

---

80. RANGE POSITION MANAGEMENT

Range:

```
50% → TP1
50% → TP2
```

TP1:

```
MidRange
```

Setelah TP1:

```
Remaining SL = actual entry price
```

BE buffer:

```
0
```

---

81. PARTIAL CLOSE

Partial volume harus mengikuti:

```
SYMBOL_VOLUME_STEP
```

Jika 50% volume tidak valid:

```
NormalizeDown(volume × 0.50)
```

Jika hasil:

```
< SYMBOL_VOLUME_MIN
```

maka partial close tidak dilakukan.

Dalam kondisi tersebut position menggunakan single TP yang valid.

---

82. BREAKOUT POSITION MANAGEMENT

Breakout:

```
100% → TP 2R
```

No trailing.

No partial close.

No BE.

---

83. TREND POSITION MANAGEMENT

Trend Pullback:

```
100% → TP 2R
```

No trailing.

No partial close.

No BE.

---

84. BREAKOUT STATE MACHINE

```
BREAKOUT_IDLE
       │
       │ valid breakout
       ▼
WAITING_RETEST
       │
       ├──── valid retest ──────► SIGNAL
       │
       ├──── failed ────────────► IDLE
       │
       └──── expired ───────────► IDLE
```

85. STATE PERSISTENCE

Breakout state harus dapat dipulihkan setelah EA restart jika terdapat active waiting setup.

State minimal:

```
BreakoutState
BreakoutDirection
BreakoutLevel
BreakoutBarTime
```

State disimpan melalui EA state mechanism.

Implementasi persistence dapat menggunakan:

```
Global Variables of Terminal
```

atau state reconstruction dari market history.

Recommendation v2.0:

```
Reconstruct from history first.
```

Global Variable hanya digunakan jika reconstruction tidak reliable.

86. RESTART BEHAVIOR

Saat `OnInit()`:

```
1. Validate symbol
2. Validate trading environment
3. Create indicator handles
4. Load current positions
5. Reconstruct strategy state
6. Initialize daily risk
7. Initialize peak equity
8. Set EA ready
```

EA tidak boleh membuka position hanya karena `OnInit()`.

Entry hanya terjadi melalui normal signal process.

---

87. MAIN EVENT LOOP

Concept:

```
OnTick():

    UpdateRuntimeState()

    PositionManager.Manage()

    if !IsNewM15Bar():
        return

    ProcessNewM15Bar()
```

---

88. NEW BAR PROCESS

```
ProcessNewM15Bar():

    UpdateMarketData()

    UpdateIndicators()

    ValidateMarketContext()

    UpdateRegime()
```

```
    UpdateMarketStructure()

    UpdateStrategyStates()

    if !Guardian.AllowNewEntry():
        return

    signal = StrategyEngine.FindSignal()

    if !signal.valid:
        return

    if !Guardian.ValidateSignal(signal):
        return

    if !RiskEngine.Calculate(signal):
        return

    if !RiskEngine.Validate(signal):
        return

    TradeExecutor.Execute(signal)
```

---

89. STRATEGY ENGINE

StrategyEngine bertugas memilih strategy berdasarkan regime.

```
switch(Regime):

    SIDEWAYS:
        RangeStrategy

    TRANSITION:
        BreakoutRetestStrategy

    TREND_BULLISH:
        TrendPullbackStrategy

    TREND_BEARISH:
        TrendPullbackStrategy
```

StrategyEngine tidak melakukan execution.

---

90. STRATEGY PRIORITY

Untuk v2.0:

```
Maximum one candidate signal per M15 candle.
```

Karena hanya satu strategy aktif berdasarkan regime, tidak terjadi strategy collision.

Jika future version memungkinkan beberapa strategy aktif dalam satu regime, StrategyEngine harus memiliki explicit priority.

---

91. LOGGING CONTRACT

Logger menyediakan level:

```
INFO
DEBUG
WARNING
ERROR
TRADE
```

Format:

```
[TIME][MODULE][LEVEL] message
```

Contoh:

```
[2026.10.06 10:15][RegimeEngine][INFO]
Regime=TREND_BULLISH ADX=25.4 Slope=0.081
```

---

92. SIGNAL LOG

Saat candidate:

```
[SIGNAL]
Strategy=TREND_PULLBACK
Direction=BUY
Entry=...
SL=...
TP=...
```

93. REJECTION LOG

Contoh:

```
[GUARDIAN][REJECT]
Reason=SpreadTooHigh
```

atau:

```
[RISK][REJECT]
Reason=VolumeBelowMinimum
```

atau:

```
[STRATEGY][NO_SIGNAL]
Reason=NoBullishConfirmation
```

94. ERROR HANDLING

Error dibagi:

```
DATA_ERROR
CALCULATION_ERROR
VALIDATION_ERROR
EXECUTION_ERROR
STATE_ERROR
```

Critical error:

```
EA tidak boleh membuka trade
```

95. DATA FAILURE

Jika:

```
CopyRates()
CopyBuffer()
SymbolInfo*
```

gagal atau data tidak cukup:

```
No signal
```

EA tidak menggunakan fallback arbitrary.

---

96. MEMORY / PERFORMANCE

EA tidak boleh:

```
Copy seluruh historical data setiap tick
```

Gunakan:

```
CopyRates()
CopyBuffer()
```

dengan jumlah data minimum yang diperlukan.

Position management dapat berjalan setiap tick.

Strategy calculation hanya pada new M15 bar.

---

97. INDICATOR HANDLE LIFECYCLE

Indicator handles dibuat pada:

```
OnInit()
```

Handle dilepas pada:

```
OnDeinit()
```

menggunakan:

```
IndicatorRelease()
```

Tidak membuat handle baru setiap tick.

---

98. TESTABILITY

Module harus dapat diuji secara terpisah.

Contoh:

```
RegimeEngine
    input data
    ↓
expected regime
```

```
RangeStrategy
    input MarketContext
    ↓
expected TradeSignal
```

```
RiskEngine
    input Equity + SL + SymbolSpec
    ↓
expected Volume
```

```
Guardian
    input runtime state
    ↓
PASS / REJECT
```

---

99. STRATEGY TEST CONTRACT

Untuk setiap strategy minimal test:

```
Valid BUY
Valid SELL
No Signal
Invalid Regime
Invalid Structure
Invalid Confirmation
Invalid SL
```

---

100. RISK TEST CONTRACT

Test:

```
Equity = 10000
Risk = 0.50%
```

Expected:

```
RiskAmount = 50
```

Test juga:

```
Large SL
Small SL
Different TickSize
Different TickValue
Different VolumeStep
VolumeMin
VolumeMax
```

---

101. NON-REPAINTING CONTRACT

EA tidak boleh:

```
Use candle 0 for historical signal
Use unconfirmed pivot
Use future candle
Recalculate past signal using future data
```

Confirmed pivot hanya digunakan setelah:

```
SwingRight candles
```

selesai.

---

102. BACKTEST MODEL

Backtest menggunakan model dengan tick accuracy terbaik yang tersedia di MT5.

Verifikasi:

```
No lookahead
No repaint
No future pivot
Correct spread
Correct commission
Correct swap
Correct symbol specification
```

---

103. STRATEGY PERFORMANCE

Performance harus dipisahkan:

```
Range
Breakout Retest
Trend Pullback
```

Dan:

```
SIDEWAYS
TRANSITION
TREND
```

Metrics:

```
Net Profit
Profit Factor
```

```
Expectancy
Win Rate
Average Win
Average Loss
Maximum Drawdown
Recovery Factor
Total Trades
Consecutive Losses
Average Trade
```

---

104. OPTIMIZATION RULE

Jangan optimize seluruh parameter sekaligus.

Urutan:

```
1. Regime
2. Range
3. Breakout
4. Pullback
5. Risk
```

Gunakan:

```
In-Sample
Out-of-Sample
Forward Test
```

Parameter yang menghasilkan maximum net profit bukan otomatis parameter terbaik.

---

105. VERSION 2.0 EXCLUSIONS

Belum digunakan:

```
M5
News Filter
AI
Machine Learning
Trailing Stop
Grid
Martingale
Averaging Down
```

```
Pyramiding
Multiple simultaneous positions
Dynamic risk
Correlation filter
Bollinger entry
Complex portfolio management
```

Architecture harus memungkinkan fitur tersebut ditambahkan kemudian tanpa mengubah strategy contract utama.

---

106. FUTURE STRATEGY ADDITION

Untuk menambahkan strategy baru:

```
Strategies/
    NewStrategy.mqh
```

Implement:

```
IStrategy
```

Kemudian register pada:

```
StrategyEngine
```

Strategy baru harus menghasilkan:

```
TradeSignal
```

dan tidak boleh memanggil:

```
CTrade
```

secara langsung.

---

107. FUTURE EXAMPLE

Misalnya ingin menambahkan:

```
LiquiditySweepStrategy
```

Architecture:

```
LiquiditySweepStrategy
        │
        ▼
    TradeSignal
        │
        ▼
     Guardian
        │
        ▼
    RiskEngine
        │
        ▼
   TradeExecutor
```

Core:

```
RiskEngine
Guardian
TradeExecutor
PositionManager
```

tidak perlu di-rewrite.

---

108. INPUT CONFIGURATION

Semua parameter strategy harus dikelompokkan.

Contoh:

```
=== GENERAL ===

MagicNumber
RiskPerTrade
MaxTotalOpenPositions

=== REGIME ===

TrendADX
```

```
SidewaysADX
TrendSlopeThreshold
SlopeLookback

=== RANGE ===

RangeLookback
MinRangeATR
MaxRangeATR
InteractionATR
RangeZonePercent
RejectionWickBody

=== BREAKOUT ===

BreakoutBodyATR
BreakoutADX
RetestATR
MaxRetestBars

=== STRUCTURE ===

SwingLeft
SwingRight

=== PULLBACK ===

PullbackMinATR
PullbackZoneATR

=== RISK ===

MaxRiskPerTrade

=== GUARDIAN ===

MaxSpreadPoints
UseSessionFilter
SessionStart
SessionEnd
DailyLossLimit
MaxDrawdown

=== DEBUG ===

DebugMode
```

---

109. CONFIGURATION RULE

Input parameter hanya digunakan jika parameter tersebut benar-benar mempunyai fungsi.

Tidak boleh membuat puluhan parameter configurable tanpa kebutuhan.

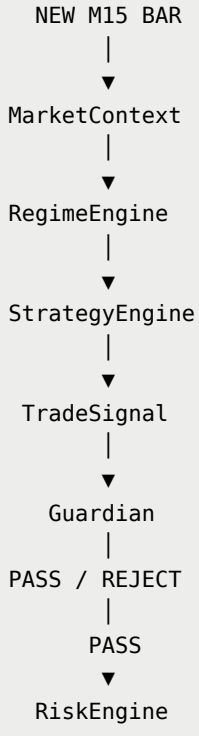
Default parameter berasal dari specification:

```
Risk = 0.50%
TrendADX = 20
SidewaysADX = 18
Slope = 0.05
Range = 20
MinRange = 1.5 ATR
MaxRange = 4 ATR
Interaction = 0.20 ATR
Rejection = 1.5x Body
BreakoutBody = 0.50 ATR
BreakoutADX = 20
Retest = 0.20 ATR
Expiry = 8 bars
Pullback = 0.30 ATR
SLBuffer = 0.20 ATR
RR = 2.0
DailyLoss = 2%
MaxDD = 10%
```

---

110. FINAL EXECUTION PIPELINE

Setiap entry wajib mengikuti:



```
                  |
          Volume Calculation
                  |
                  ▼
          Final Validation
                  |
                  ▼
          TradeExecutor
                  |
                  ▼
               CTrade
                  |
                  ▼
              Broker
                  |
                  ▼
          PositionManager
```

---

111. CRITICAL SEPARATION

Rule yang wajib dipertahankan:

```
STRATEGY
"What should we trade?"

GUARDIAN
"Are we allowed to trade?"

RISK ENGINE
"How much can we trade?"

EXECUTOR
"How do we send the trade?"

POSITION MANAGER
"How do we manage an existing trade?"
```

Tidak boleh dicampur.

---

112. DEFINITION OF DONE

EA v2.0 dianggap implementation-ready jika:

```
[ ] Semua indicator memiliki handle lifecycle
[ ] Semua signal menggunakan closed candle
[ ] Tidak ada lookahead
```

```
[ ] Pivot hanya confirmed pivot
[ ] Regime deterministic
[ ] Range deterministic
[ ] Breakout deterministic
[ ] Retest deterministic
[ ] Retest expiry deterministic
[ ] Pullback deterministic
[ ] TradeSignal standard
[ ] Strategy tidak melakukan order
[ ] RiskEngine menghitung volume
[ ] Guardian melakukan entry validation
[ ] TradeExecutor satu-satunya module yang execute order
[ ] PositionManager mengelola existing position
[ ] MagicNumber filtering
[ ] Symbol filtering
[ ] Broker specification validation
[ ] Stop level validation
[ ] Volume normalization
[ ] Price normalization
[ ] Daily loss protection
[ ] Drawdown protection
[ ] Spread protection
[ ] Session protection
[ ] Partial close Range
[ ] Break-even Range
[ ] State recovery
[ ] Error logging
[ ] Debug logging
[ ] Backtest reproducible
```

---

113. IMPLEMENTATION ORDER

Implementasi tidak dilakukan sekaligus.

Phase 1 — Core

```
Types
Config
MarketContext
TradeSignal
EAState
Logger
```

Phase 2 — Market

```
MarketData
IndicatorManager
RegimeEngine
MarketStructure
```

Phase 3 — Strategy Interface

```
IStrategy
StrategyEngine
```

Phase 4 — Range

```
RangeStrategy
```

Test sampai valid.

Phase 5 — Breakout

```
BreakoutStrategy
BreakoutState
Retest
Expiry
Failure
```

Test sampai valid.

Phase 6 — Trend Pullback

```
TrendPullbackStrategy
```

Test sampai valid.

Phase 7 — Risk

```
RiskEngine
Volume calculation
```

```
Broker specification
```

Phase 8 — Safety

```
Guardian
Daily loss
Drawdown
Spread
Session
Portfolio
```

Phase 9 — Execution

```
TradeExecutor
CTrade
```

Phase 10 — Position Management

```
PositionManager
Break-even
Partial close
```

Phase 11 — Integration

```
Full EA
```

Phase 12 — Backtest

```
Functional testing
Regression testing
Strategy testing
Risk testing
Robustness testing
```

---

114. FINAL ARCHITECTURE

```
TradingEA.mq5
│
├── Core
│   ├── Types
│   ├── Config
│   ├── MarketContext
│   ├── TradeSignal
│   └── EAState
│
├── Market
│   ├── MarketData
│   ├── IndicatorManager
│   ├── RegimeEngine
│   └── MarketStructure
│
├── Strategies
│   ├── IStrategy
│   ├── StrategyEngine
│   ├── RangeStrategy
│   ├── BreakoutStrategy
│   └── TrendPullbackStrategy
│
├── Risk
│   ├── Guardian
│   └── RiskEngine
│
├── Trading
│   ├── PortfolioManager
│   ├── TradeExecutor
│   └── PositionManager
│
└── Utils
    ├── PriceUtils
    ├── VolumeUtils
    ├── TimeUtils
    └── Logger
```

Core dependency:

```
Market
   ↓
Strategy
   ↓
Signal
   ↓
Guardian
   ↓
Risk
   ↓
```

```
Execution
   ↓
Position Management
```

Tidak ada dependency terbalik.

---

115. FINAL PRINCIPLE

EA v2.0 harus dibangun dengan prinsip:

```
Stable Core
+
Replaceable Strategies
+
Standard TradeSignal
+
Centralized Risk
+
Centralized Execution
+
Deterministic Market Data
```

Dengan architecture ini:

```
Tambah strategy
        ↓
Tambah 1 module
        ↓
Register ke StrategyEngine
```

bukan:

```
Tambah strategy
        ↓
Modify OnTick()
Modify Risk
Modify CTrade
Modify PositionManager
Modify existing strategies
Modify state
```

Target akhir adalah:

```
Trading Logic
    ≠
Execution Logic
    ≠
Risk Logic
    ≠
Position Management
```

sehingga EA dapat berkembang secara incremental tanpa menjadi satu file besar yang sulit dipelihara.